SLR-OB-22

Seat No.	

Set P

B.Sc. (E.C.S.) (Semester - III) (New) (NEP CBCS) Examination: March/April – 2026
Operating Systems (G06-GE-OE-0301)

Day & Date: Thursday, 02-04-2026 Max. Marks: 30
Time: 09:00 AM To 10:30 AM

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose correct alternative. (MCQ) **06**

1) Which of the following is NOT a type of Operating System?
a) Batch b) Multiprogramming
c) Real-Time d) BIOS

2) Which scheduling algorithm is preemptive by nature?
a) First Come First Serve (FCFS)
b) Shortest Job First (SJF)
c) Round Robin (RR)
d) Priority Scheduling (non-preemptive)

3) The main function of a Process Control Block (PCB) is: ____.
a) To store program source code
b) To store process-related information
c) To store hardware details
d) To schedule jobs

4) Dining Philosophers problem can NOT be solved using Two Process solutions.
a) TRUE b) FALSE

5) Which synchronization tool is used to solve the critical section problem?
a) System calls b) Semaphores
c) Context switch d) PCB

6) ____ provides the interface between a running program and O.S.
a) System Process b) Overlays
c) Kernel d) system calls

Q.2 Answer the following. (Any Three) **06**

a) Define an Operating System. List out different types of OS.
b) Define system calls with two examples.
c) Define process states with a neat diagram.
d) Define context switching with its Drawback.
e) List out OS Services.

Q.3 Answer the following. (Any Two) **06**

a) Explain Multiprogramming OS.
b) Explain the Dining Philosopher Problem.
c) Write a note on Multilevel Queue Scheduling with a neat diagram.

Q.4 Answer the following. (Any Two) **06**

a) Explain the Producer Consumer Problem with a diagram.
b) Explain the Critical Section Problem and the role of semaphores.
c) State Scheduling Criteria's for Scheduling algorithms.

Q.5 Answer the following. (Any One) **06**

a) Define Process Scheduling and Consider Following System Snapshot,

Process	**P1**	**P2**	**P3**	**P4**	**P5**
Arrival Time	0	1	2	3	4
CPU Burst	5	9	7	2	4

Prepare Gantt chart and calculate Average Waiting and Average Turnaround Time using, RR Scheduling Algorithm with Time Slice= 2 m/s

b) Explain PCB with neat diagram.